

MSK Path Disease Chart

Every bone, joint, soft tissue, nerve and muscle disease in the MSK pathology block — pathogenesis, who and where, the key findings, and the pearl most likely to be tested. Yellow column = high yield.

Updated September 30, 2026 · Revision 1.0 · 82 entities

■ Bone Disorders ■ Benign Bone Tumors ■ Malignant Bone Tumors ■ Soft Tissue ■ Peripheral Nerve & Muscle ■ Rheumatology

STEP 1 board topic beyond the in-house block

Bone Disorders Genetic, metabolic, fracture, vascular and infectious disease of bone · 14 shown

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
GENETIC / DEVELOPMENTAL				
Osteogenesis imperfecta Brittle bone disease MSK10	COL1A1/COL1A2 mutation → defective or ↓ type I collagen (mostly AD; rare AR: CRTAP, LEPRE1). Type I = ↓ amount of normal collagen (mildest, most common); II perinatal lethal; III progressive deforming; IV moderate, normal sclerae.	Child/young adult with multiple low-trauma fractures, blue sclerae , conductive + sensorineural hearing loss, dentinogenesis imperfecta, joint laxity, scoliosis; family history of "osteoporosis"; very low Z-score.	X-ray: diffuse osteopenia, multiple fractures of varying age, bowing, Wormian bones . Labs normal (Ca ²⁺ , phosphate, PTH, ALP). Thin sclera shows underlying choroid → blue. Teeth gray-blue/yellow-brown, fragile.	"I can't BITE ": Bones, I (eyes), Teeth, Ears. Trap: multiple fractures mimic child abuse → look for blue sclerae, teeth, hearing loss. Type I collagen defect in bone, skin, sclera, dentin.
Achondroplasia MSK10	FGFR3 gain-of-function (constitutive activation) → inhibits chondrocyte proliferation in growth plate → impaired endochondral ossification; membranous ossification normal. AD, most cases sporadic (↑ paternal age). Related FGFR3 mutation → thanatophoric dysplasia .	Most common skeletal dysplasia/dwarfism: short limbs, normal-length trunk, macrocephaly, frontal bossing, saddle nose. Normal intelligence, lifespan, fertility. Thanatophoric : very short limbs, small chest → lethal pulmonary hypoplasia at birth.	Short, thick long bones; narrowed growth plates with disorganized chondrocytes; normal bone density; foramen magnum/spinal stenosis. Labs normal. Thanatophoric: bell-shaped narrow thorax, telephone-receiver femurs.	Not a collagen disease: normal bone density , normal trunk. FGFR3 activation = brakes on growth plate. Most common cause of dwarfism; de novo mutation, ↑ paternal age. Thanatophoric = most common lethal skeletal dysplasia.
Osteopetrosis Marble bone disease MSK10	Defective osteoclast resorption : often carbonic anhydrase II mutation → cannot acidify resorption pit; or RANKL-pathway activation defects. AR = severe infantile; AD = mild, found later in adults.	Infant (AR): pancytopenia, infections, hepatosplenomegaly (extramedullary hematopoiesis), cranial nerve palsies (vision, hearing, facial) from narrowed foramina. Adult (AD): incidental dense bones, fractures. Bones dense but brittle.	X-ray: diffuse sclerosis, " bone-in-bone " appearance. Histology: persistent primary spongiosa (calcified cartilage cores), woven bone without lamellar bone, no marrow space . Labs usually normal (↓ Ca ²⁺ in severe AR).	Petrified = hard but shatters. Dense bone + anemia + HSM + CN palsies . Severe infantile form cured by bone marrow transplant (osteoclasts derive from monocytes). CA II defect also → renal tubular acidosis.
METABOLIC				
Osteoporosis (primary) Postmenopausal / senile MSK10	↓ Bone mass with normal mineralization; resorption > formation. Postmenopausal : estrogen loss → ↑ RANKL, ↓ OPG, ↑ IL-6. Senile : aging osteoblasts, fewer growth factors. Low peak bone mass + fast loss. Trabecular bone hit first.	Postmenopausal woman or elderly: often asymptomatic until fracture. Vertebral compression (anterior wedge) → back pain, height loss, kyphosis; hip fracture (↑ 1-yr mortality, nursing care); distal radius (Colles) fracture.	Labs all normal (Ca ²⁺ , phosphate, PTH, ALP). DXA T-score ≤ -2.5 or fragility fracture = diagnosis. Histology: thin, disconnected trabeculae; cortical thinning. X-ray: osteopenia, wedge fractures.	Normal labs + fragility fracture. Cancellous/trabecular bone first → vertebrae. Peak bone mass by 20s–30s; weight-bearing exercise + Ca ²⁺ /vit D raise it. Tx: bisphosphonates, denosumab (anti-RANKL), teriparatide (intermittent PTH).

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
Secondary osteoporosis MSK10	Glucocorticoids (↓ osteoblastogenesis, ↑ RANKL, ↓ gut Ca ²⁺), hyperthyroid, hyperparathyroid, hypogonadism, Cushing, GH deficiency; celiac, bariatric surgery, liver disease; multiple myeloma; renal osteodystrophy; aromatase inhibitors, anticoagulants (heparin), anticonvulsants; immobility.	Any age/sex with underlying disease or drug exposure: young man or premenopausal woman with fragility fracture → search for cause. Classic: celiac or long-term prednisone patient with vertebral wedge fracture.	DXA low; Z-score (age-matched) useful in young patients. Primary-disease labs abnormal (cortisol, TSH, PTH, SPEP for myeloma, tTG for celiac). Thin trabeculae as in primary.	Fragility fracture in a man or young adult = workup for secondary cause. Glucocorticoids = most common drug cause. Myeloma: lytic lesions + osteoporosis, ↑ Ca ²⁺ . Immobilization → rapid bone loss (↓ mechanical loading).
Scurvy Vitamin C deficiency MSK10	Vitamin C = cofactor for prolyl and lysyl hydroxylase (RER) → no hydroxylation → unstable triple helix, poor cross-linking → weak collagen, fragile vessels, defective osteoid. Risk: elderly with poor diet, alcoholics, malabsorption, fad diets, dialysis.	Elderly man on frozen dinners/tea-and-toast: bleeding gums , perifollicular purpura, corkscrew hairs, poor wound healing, hemarthrosis, anemia. Children: bowed long bones, depressed sternum, outward-projecting ribs (cartilage overgrows into marrow).	X-ray: ground-glass osteopenia, subperiosteal hemorrhage . Ca ²⁺ , phosphate, PTH, ALP normal. Anemia (bleeding, ↓ iron absorption). Low plasma/leukocyte ascorbate.	Matrix (collagen) defect, not mineralization. Buzzwords: perifollicular hemorrhage, corkscrew hairs, swollen bleeding gums . Vit C also needed for iron absorption and dopamine → norepinephrine. Trap: normal bone labs.
Rickets / osteomalacia MSK10	Failure to mineralize osteoid : ↓ vitamin D (low sun, dark skin, malabsorption—celiac, bariatric, biliary/pancreatic; liver/renal disease; 1α-hydroxylase defect; P450 inducers phenytoin, rifampin) or phosphate loss (antacids, X-linked hypophosphatemic —FGF23 excess).	Rickets (child, open plates): bowed legs, rachitic rosary, craniotabes, frontal bossing, pectus carinatum, waddling gait. Osteomalacia (adult): bone pain, proximal muscle weakness, fractures.	Labs: ↓ 25-OH D, ↓/N Ca ²⁺ , ↓ phosphate , ↑ PTH , ↑ ALP . Rickets X-ray: cupping and fraying of metaphyses ; thick growth plate, lost chondrocyte columns. Osteomalacia: wide osteoid seams, Looser zones (pseudofractures).	Same defect: rickets = child, osteomalacia = adult. ↓ vit D → ↓ Ca ²⁺ → ↑ PTH → renal phosphate wasting → ↓ Ca×P. XLH: normal Ca ²⁺ , ↓ phosphate, ↓ 1,25-D. Anticonvulsants → osteomalacia.
Hyperparathyroid bone disease Osteitis fibrosa cystica / brown tumor MSK10	Continuous PTH → ↑ RANKL on osteoblasts → osteoclast resorption, especially cortical bone . Primary: parathyroid adenoma (most) or hyperplasia. Secondary: low Ca ²⁺ /vit D, CKD. Tertiary: autonomous glands after long secondary HPT.	Primary: often incidental hypercalcemia; "stones, bones, groans, moans" —kidney stones, fractures/bone pain, abdominal pain, "brain fog". Bone loss at distal forearm/wrist. Tertiary: dialysis patient becomes hypercalcemic.	Primary: ↑ Ca²⁺ , ↓ phosphate , ↑ PTH , ↑ urinary cAMP. X-ray: subperiosteal resorption of radial middle phalanges , distal clavicle; salt-and-pepper skull. Histo: dissecting osteitis ; brown tumor = giant cells + hemorrhage + fibrosis.	"Railroad tracks" = osteoclasts tunneling through trabecula centers. Brown tumor is brown from hemosiderin (not a neoplasm). Sestamibi localizes adenoma. Trap: brown tumor mimics giant cell tumor—check Ca ²⁺ /PTH.
Renal osteodystrophy MSK10	Bone disease of CKD : ↓ 1α-hydroxylase → ↓ 1,25-D → ↓ Ca ²⁺ ; phosphate retention (↓ excretion, ↑ FGF23) → secondary hyperparathyroidism . Mixed osteitis fibrosa, osteomalacia, adynamic bone; secondary osteoporosis cause.	Dialysis/CKD patient with bone pain, fractures, pruritus, vascular/soft-tissue calcification. Long-standing disease → tertiary HPT (autonomous glands, hypercalcemia). Children: growth failure, rickets-like changes.	Labs: ↓/N Ca ²⁺ , ↑ phosphate , ↑ PTH , ↑ ALP , ↓ 1,25-D. X-ray: subperiosteal resorption, "rugger jersey" spine (banded sclerosis), brown tumors. Histo: osteitis fibrosa ± wide osteoid seams.	Key contrast: secondary HPT of CKD has ↑ phosphate ; vit D—deficiency secondary HPT has ↓ phosphate. Tx: phosphate binders, calcitriol, cinacalcet. Dialysis patient now hypercalcemic = tertiary HPT .
Paget disease of bone Osteitis deformans MSK10	Localized disordered remodeling: osteoclast precursors hypersensitive to vit D and RANKL; IL-6 recruits more. Possible paramyxovirus (measles, RSV) + genetics (SQSTM1). Phases: lytic → mixed → sclerotic (burnt out). Net ↑ bone mass but weak.	Adult >40–50, Northern European descent; often asymptomatic. Skull, spine, pelvis, femur/tibia: bone pain, warmth, ↑ hat size , hearing loss, bowed tibia, spinal stenosis, fractures, high-output heart failure (AV shunting), osteosarcoma .	Isolated ↑ ALP ; Ca ²⁺ , phosphate, PTH normal. X-ray: cotton-wool skull , mixed lytic/sclerotic, thick cortex; hot bone scan. Histo: mosaic lamellar bone with irregular cement lines; giant osteoclasts (up to ~100 nuclei); fibrovascular marrow.	Isolated ↑ ALP in older adult = Paget until proven otherwise. Mosaic pattern = favorite image. Never in children. Complications: osteosarcoma, deafness, high-output HF. Tx: bisphosphonates .
FRACTURE & VASCULAR				

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
Fracture & fracture healing MSK11	Healing: hematoma (0–1 d; PDGF, TGF-β, FGF) → granulation + soft callus of fibro/hyaline cartilage (≤2 wk; BMPs, VEGF) → bony callus of woven bone by endochondral ossification (2–3 wk) → remodeling to lamellar bone (months).	Terms: complete/incomplete (greenstick), closed/ compound (open → osteomyelitis), comminuted (fragments), displaced, pathologic (diseased bone, minimal trauma), stress (repetitive load—runners, recruits). Assess vascular, neuro, soft tissue first.	X-ray confirms type; callus visible ~2–3 wk. Histo: callus = cartilage + woven bone . ↑ ALP during healing. Nonunion/ pseudarthrosis with motion; malunion with displacement/comminution.	Compound = comes out the skin ; comminuted = crumbs. Woven bone in adult = pathologic/reparative. Needs blood supply + immobilization; diabetes, steroids, infection delay healing. Femoral neck/scaphoid → AVN.
Avascular necrosis Osteonecrosis MSK11	Ischemic death of bone + marrow; end-arterial subchondral supply. Causes " ASEPTIC ": Alcohol, Sickle cell, Exogenous steroids , Pancreatitis, Trauma (femoral neck fracture—medial circumflex), Idiopathic (Legg-Calvé-Perthes), Caisson. Also radiation, Gaucher, lupus, pregnancy.	Classic femoral head ; also humeral head, femoral condyles, scaphoid, talus. Lupus patient on prednisone, alcoholic, sickle cell, deep-sea diver with hip/groin pain on movement → later at rest → collapse → secondary OA .	MRI most sensitive (X-ray normal early); later crescent sign , collapse. Gross: pale yellow wedge-shaped subchondral infarct. Histo: empty lacunae , necrotic marrow, creeping substitution at edges.	Empty lacunae = dead bone. Articular cartilage survives (fed by synovial fluid diffusion) while bone collapses. Glucocorticoids = very common cause. Legg-Calvé-Perthes: boy 4–10 with limp. Often ends in joint replacement.
INFECTION				
Pyogenic osteomyelitis MSK11	Staph aureus (80–90%). Routes: hematogenous (children), contiguous (diabetic foot), direct inoculation (open fracture). Special: sickle cell → Salmonella ; IVDU/sneaker puncture → Pseudomonas; neonate → GBS, E. coli; cat/dog bite → Pasteurella.	Children: metaphysis of long bone (slow looping capillaries); neonates: epiphysis/joint; adults: vertebrae, subchondral bone. Fever, local pain. Chronic (5–25%): draining sinus, pathologic fracture, amyloidosis, sepsis, SCC in sinus tract .	MRI most sensitive ; X-ray normal early. Blood + bone cultures; ↑ ESR/CRP. Acute: neutrophils, necrotic bone, subperiosteal abscess. Chronic: sequestrum + involucrum . Brodie abscess : lucency + sclerotic rim.	Sequestrum = stiff dead bone; Involucrum = new bone shell. Sickle cell → Salmonella (S. aureus still most common overall). Draining sinus → squamous cell carcinoma (Marjolin-type). ESR/CRP track treatment.
Tuberculous osteomyelitis Pott disease MSK11	Hematogenous spread of M. tuberculosis from visceral (usually lung) focus; destructive, treatment-resistant. Nearly half involve the spine (thoracic/lumbar). Caseating granulomatous inflammation destroys vertebral bodies and discs.	Back pain, weight loss, night sweats, from lung focus. Vertebral collapse → gibbus deformity (kyphosis), compression fractures, cord compression/neurologic deficits; cold psoas abscess tracking to groin.	X-ray/MRI: vertebral body destruction with disc space involvement , adjacent vertebrae, paraspinal/psoas abscess. Histo: caseating granulomas , Langhans giant cells; AFB stain/culture.	Vertebral destruction + disc involvement + psoas abscess = Pott disease . Contrast: pyogenic vertebral osteomyelitis in IVDU (S. aureus, Pseudomonas). Gibbus = sharp angular kyphosis.

Benign Bone Tumors

Bone-forming, cartilage-forming and tumor-like lesions · 10 shown

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
BONE-FORMING				
Osteoma MSK16	Benign outgrowth of dense mature lamellar bone. Multiple osteomas → think Gardner syndrome (AD FAP variant, APC loss → β-catenin accumulation) with colon polyps, desmoids, epidermal cysts, dental anomalies, CHRPE.	Middle-aged adults; skull, facial bones, paranasal sinuses, jaw (mandible). Usually asymptomatic hard, slow-growing mass; may obstruct sinus drainage or cause cosmetic deformity. No malignant transformation.	X-ray: well-defined, dense, homogeneous radiopaque surface mass projecting from bone. Micro: compact mature lamellar (± woven) bone with few cells.	Jaw/skull osteomas + colon polyps + desmoids + supernumerary teeth = Gardner syndrome . Trap: find osteomas → screen colon (near-certain colorectal cancer without colectomy).
Osteoid osteoma MSK14	Benign osteoblastic tumor; nidus osteoblasts make abundant prostaglandin E2 → intense pain and reactive sclerosis. Same histology as osteoblastoma, distinguished by size (<2 cm).	Teens/young adults (10–20s), M>F; cortex of long-bone diaphysis (femur, tibia) near the surface. Severe night pain relieved by NSAIDs/aspirin . Benign; excision or radiofrequency ablation curative.	X-ray/CT: small (<2 cm) round lucent nidus surrounded by thick dense sclerotic cortical rim. Micro: interlacing woven-bone trabeculae rimmed by osteoblasts in vascular stroma.	Young patient + night pain + relieved by aspirin/NSAIDs + lucent nidus with sclerotic rim = osteoid osteoma. Trap: >2 cm, spine, NOT relieved by NSAIDs = osteoblastoma.
Osteoblastoma MSK14	Benign osteoblastic tumor histologically identical to osteoid osteoma but larger (>2 cm); less prostaglandin → less reactive sclerosis and less intense pain.	Adolescents/young adults; posterior elements of the vertebrae (lamina, pedicle, spinous process). Dull aching pain not relieved by NSAIDs ; may cause scoliosis or nerve compression. Can recur locally.	X-ray: expansile lytic lesion >2 cm, little surrounding sclerosis, ± mineralized center. Micro: woven bone trabeculae rimmed by prominent osteoblasts; no permeation of host bone (vs osteosarcoma).	Osteoblastoma = bigger, back (posterior spine), NSAIDs don't help. Osteoid osteoma = small, long-bone cortex, NSAIDs help. Trap: mimic of osteosarcoma — lacks infiltration and atypia.
CARTILAGE-FORMING				
Osteochondroma Exostosis MSK14	Most common benign bone tumor. Loss of both copies of EXT1/EXT2 in growth-plate chondrocytes → displaced growth plate. Multiple hereditary exostoses: AD, EXT, childhood; small secondary chondrosarcoma risk.	Adolescents/young adults; metaphysis of long bones near the knee, growing away from the joint . Hard, sometimes tender painless lump. Benign; stops growing at skeletal maturity; excise if symptomatic.	X-ray/gross: pedunculated or sessile bony stalk with cortex and marrow continuous with parent bone . Micro: hyaline cartilage cap (1–2 cm, purple) → endochondral ossification → cancellous bone (pink).	Most common benign bone tumor; teen + metaphyseal stalk continuous with marrow + cartilage cap. Trap: cap >2 cm, growth or pain after skeletal maturity → suspect secondary chondrosarcoma.
Enchondroma MSK14	Benign intramedullary hyaline cartilage tumor; IDH1/IDH2 mutations. Multiple = Ollier disease (enchondromatosis); Maffucci syndrome = enchondromas + soft tissue hemangiomas. Both ↑ risk of chondrosarcoma.	Adults 20–50; medullary metaphysis of small bones of hands and feet (phalanges). Usually asymptomatic or pathologic fracture. Peripheral cartilage lesions almost always benign; same lesion centrally suspicious.	X-ray: well-circumscribed central lucency, thin sclerotic rim, ring-and-arc (popcorn) calcification, intact cortex. Micro: lobules of bland hyaline cartilage, low cellularity, no atypia.	Cartilage in a phalanx = enchondroma (benign). Ollier = multiple enchondromas; Maffucci = Ollier + hemangiomas . Trap: pain or growth in pelvic/central cartilage lesion → chondrosarcoma.
OTHER / TUMOR-LIKE				
Giant cell tumor Osteoclastoma MSK14	Neoplastic mononuclear stromal cells (H3F3A mutation) overexpress RANKL → recruit and activate osteoclast-like giant cells → bone lysis. Benign but locally aggressive; rare malignant GCT.	Adults 20–40 (closed growth plate); epiphysis extending to metaphysis, around the knee (distal femur, proximal tibia). Pain, swelling, pathologic fracture. Recurrs after curettage ; rare benign lung implants.	X-ray: eccentric, expansile lytic soap bubble lesion reaching subchondral bone. Micro: multinucleated osteoclast-like giant cells evenly spread among plump mononuclear stromal cells whose nuclei match the giant-cell nuclei.	Grown-ups at the End: 20–40 + epiphysis + soap bubble = GCT. Trap: "benign" yet locally aggressive and recurs. Denosumab (anti-RANKL) treats unresectable disease.

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
Aneurysmal bone cyst ABC MSK14	Benign, locally destructive osteoclastic giant cell-rich lesion; primary ABC driven by USP6 rearrangements (CDH11-USP6); secondary ABC change can occur in other tumors (GCT, osteoblastoma).	Children/adolescents (<20); metaphysis of long bones and posterior vertebral elements. Pain and swelling, rapid expansion, pathologic fracture. Benign; curettage, may recur.	X-ray: eccentric expansile lytic, multiloculated lesion with thin bony shell; MRI fluid-fluid levels . Gross: spongy blood-filled cysts. Micro: septa of fibroblasts, osteoclast-like giant cells, reactive woven bone.	Blood-filled multiloculated cyst + fluid–fluid levels + teen metaphysis = ABC. Trap: telangiectatic osteosarcoma mimics ABC — look for malignant cells in septa.
Fibrous dysplasia MSK14	Developmental arrest: marrow replaced by fibrous tissue and immature woven bone. Activating GNAS (Gso) mutation → ↑ cAMP. Mono- or polyostotic; McCune-Albright = polyostotic + café-au-lait (coast of Maine) + precocious puberty/endocrinopathy.	Monostotic: adolescents/young adults, ribs, femur, tibia, craniofacial; often incidental. Polyostotic: childhood onset, deformity, pain, fractures. Usually diaphysis/metaphysis medullary; benign, rare sarcomatous change.	X-ray: well-defined medullary ground-glass lesion; shepherd's crook proximal femur. Micro: curvilinear woven bone trabeculae ("Chinese letters") without osteoblastic rimming in bland fibroblastic stroma.	Ground glass + shepherd's crook + Chinese letter trabeculae. McCune-Albright : GNAS mosaic (postzygotic) mutation; irregular "coast of Maine" café-au-lait (vs smooth "coast of California" in NF1).
Non-ossifying fibroma Fibrous cortical defect MSK14	Common developmental benign fibrohistiocytic lesion (not true neoplasm); small form = fibrous cortical defect, larger (>5 cm) = NOF. Multiple lesions seen in Jaffe-Campanacci syndrome / NF1.	Children and adolescents; metaphysis of long bones, eccentric in cortex, distal femur/proximal tibia. Usually asymptomatic incidental finding; large lesions may fracture. Most regress spontaneously after skeletal maturity.	X-ray: eccentric, well-defined, lobulated metaphyseal lucency with thin sclerotic scalloped rim , cortex-based. Micro: storiform bland fibroblasts, scattered giant cells, foamy (lipid-laden) macrophages, hemosiderin.	Incidental eccentric sclerotic-rimmed metaphyseal lucency in a child = NOF, "don't touch" lesion. Trap: foamy macrophages + giant cells ≠ GCT (child, metaphysis, cortex-based).
Simple (unicameral) bone cyst UBC STEP 1	Benign fluid-filled cavity of unknown cause (possibly venous obstruction/impaired drainage at growth plate); not neoplastic. Clear serous fluid lined by thin fibrous membrane.	Children/adolescents 5–15, boys; metaphysis of proximal humerus (and proximal femur) adjacent to growth plate. Asymptomatic until pathologic fracture . Benign; may heal after fracture; recurs after curettage in younger kids.	X-ray: central, well-defined, unilocular lucency with thin cortex, no periosteal reaction; fallen fragment sign (cortical fragment settles in fluid). Micro: thin fibrous wall, scattered giant cells.	Child + proximal humerus + pathologic fracture + fallen fragment sign = unicameral cyst. Trap: central, unilocular, serous (vs ABC: eccentric, multiloculated, blood-filled with fluid–fluid levels).

Malignant Bone Tumors

Primary sarcomas, metastases and marrow tumors · 5 shown

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
PRIMARY				
Osteosarcoma Osteogenic sarcoma MSK14	Most common primary malignant bone tumor. RB1 (hereditary retinoblastoma, ~1000× risk) and TP53 (Li-Fraumeni), CDKN2A; MDM2/CDK4 amplification in low-grade types; MYC amplification = worse. Secondary: Paget disease, bone infarcts, radiation.	Bimodal: 75% <20 (teen boys), 25% elderly with Paget/radiation. Metaphysis around the knee (distal femur, proximal tibia ~50%). Painful swelling, pathologic fracture. Hematogenous lung mets; neoadjuvant chemo + limb-sparing surgery, ~70% survival.	X-ray: destructive mixed lytic-blastic mass, Codman triangle (lifted periosteum), sunburst . Gross: gray-white gritty hemorrhagic mass breaking cortex. Micro: pleomorphic malignant cells directly making lace-like osteoid ; telangiectatic subtype mimics ABC.	Teen knee metaphysis + Codman triangle + sunburst + malignant osteoid. Trap: elderly with Paget disease + new pain/mass = secondary osteosarcoma. Retinoblastoma survivor → osteosarcoma.
Chondrosarcoma MSK14	Second most common primary malignant bone tumor; malignant cartilage only (no osteoid). IDH1/IDH2 mutations; arises from enchondromas in Ollier/Maffucci or from osteochondroma cap (EXT, multiple hereditary exostoses).	Adults >45 (peak 60s–70s); central/axial skeleton : pelvis (ilium), proximal femur, humerus, shoulder girdle, ribs. Deep dull pain worse at night. Grade 1–3 predicts outcome ; high grade → lung mets. Chemo/radiation resistant; surgery.	X-ray: lytic with ring-and-arc (popcorn) calcification , endosteal scalloping, cortical thickening. Gross: lobulated glistening gray-blue translucent ("bubbly") nodules. Micro: atypical chondrocytes in lacunae, binucleation, permeation of host bone.	Older adult + pelvis/shoulder + growing painful cartilage mass = chondrosarcoma. Trap: identical bland cartilage is benign in phalanx (enchondroma) but malignant in pelvis. Grade = prognosis.
Ewing sarcoma MSK14	Primitive small round cell sarcoma (neuroectodermal). t(11;22) EWSR1-FLI1 (~85–90%) → aberrant fusion oncoprotein; t(21;22) EWSR1-ERG alternate. Simple translocation karyotype; high grade by definition.	Children/teens 5–20, boys, White > Black; diaphysis of long bones (femur) and pelvis . Painful enlarging mass with fever, ↑ ESR, leukocytosis → mimics osteomyelitis. Aggressive but chemosensitive ; mets to lung/bone.	X-ray: moth-eaten/permeative diaphyseal lesion with onion-skin layered periosteal reaction, soft tissue mass. Micro: sheets of uniform small round blue cells , PAS+ glycogen, Homer-Wright rosettes. IHC: CD99 (membranous), NKX2.2, FLI1.	Child + diaphysis + onion skin + fever = Ewing (11 + 22 = 33, Patrick Ewing's jersey). Trap: febrile kid with lytic diaphyseal lesion isn't always osteomyelitis — biopsy; CD99 + EWSR1 rearrangement confirms.
SECONDARY / MARROW				
Metastatic carcinoma to bone MSK14	Most common malignancy of bone (>> primary). Hematogenous to red marrow (Batson plexus to spine). Primaries: prostate, breast , lung, kidney, thyroid. Lytic via PTHrP/RANKL osteoclast activation; blastic via tumor-driven osteoblasts.	Adults >40; multifocal, axial skeleton (vertebrae, pelvis, ribs, skull) and proximal limbs; rare distal to knee/elbow (hand mets ~half from lung). Bone pain, pathologic fracture, hypercalcemia, cord compression; often first sign of cancer.	X-ray: blastic = prostate; lytic = RCC, thyroid (and myeloma); mixed = breast, lung. Bone scan hot (except purely lytic). Micro: carcinoma (glands, keratin+) in marrow; IHC identifies primary (PSA, TTF-1, PAX8, GATA3).	Multiple bone lesions in an adult = metastasis until proven otherwise. "Permanently Branded Lungs Kill The patient." Prostate → blastic, ↑ ALP; RCC/thyroid → lytic, often pulsatile/vascular.
Multiple myeloma Plasma cell myeloma MSK14	Monoclonal plasma cell neoplasm of marrow. Myeloma cells secrete IL-6, RANKL, MIP-1α, DKK1 → osteoclast activation + osteoblast suppression → purely lytic lesions. M protein (IgG > IgA), light chains (Bence Jones); preceded by MGUS.	Older adults (~70), Black > White; axial skeleton : vertebrae, skull, ribs, pelvis. CRAB : hyperCalcemia, Renal failure (cast nephropathy), Anemia, Bone pain/fractures; recurrent infections. AL amyloidosis possible.	X-ray: punched-out lytic lesions (skull "raindrop"), diffuse osteopenia; bone scan often cold, ALP normal . SPEP M spike; UPEP Bence Jones. Marrow >10% clonal plasma cells (CD138+, clock-face chromatin); smear rouleaux .	Elderly + back pain + hypercalcemia + renal failure + anemia + punched-out skull lesions = myeloma. Trap: normal ALP and cold bone scan (no osteoblast activity) despite extensive bone destruction.

Soft Tissue

Fibrous, fatty, muscle, vascular and nerve-sheath tumors · 14 shown

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
FIBROUS				
Nodular fasciitis MSK16	Benign, self-limited reactive (pseudosarcomatous) myofibroblastic proliferation, often after trauma. Actually clonal: t(17;22) MYH9-USP6 fusion (transient neoplasm).	Young adults 20–40; subcutis of volar forearm (also trunk, head/neck). Small (1–3 cm), sometimes tender nodule that grows rapidly over weeks . Benign; excision curative, does not recur ; may regress.	Micro: plump uniform myofibroblasts in loose, feathery tissue culture–like pattern with microcysts, extravasated RBCs , scattered lymphocytes; mitoses frequent but not atypical . IHC: SMA+, desmin–.	Rapidly growing forearm nodule in a young adult weeks after trauma + tissue-culture fibroblasts + extravasated RBCs = nodular fasciitis. Trap: mitoses and fast growth mimic sarcoma — reassure.
Superficial fibromatosis Dupuytren (palmar), Ledderhose (plantar), Peyronie (penile) MSK16	Benign infiltrative fibroblastic proliferation of fascia/aponeuroses. Associations: Northern European men, alcohol use, diabetes, smoking, epilepsy (antiepileptics), family history. Some share β -catenin (Wnt) pathway alterations.	Adults (older men); palmar fascia → flexion contracture of 4th/5th fingers (Dupuytren); non–weight-bearing plantar sole (Ledderhose); dorsolateral penile shaft plaque → curvature (Peyronie). Slow, small; recurs; never metastasizes.	Gross: firm nodules and cords in palmar/plantar fascia. Micro: bland fibroblasts/myofibroblasts in nodules with abundant dense collagen; proliferative (cellular) → involutional (collagen-rich) phases; no atypia.	Palmar cord + ring/little finger flexion contracture = Dupuytren. Trap: fibromatoses are "benign" yet infiltrative and recurrent; superficial types are small and slow, deep (desmoid) larger and faster.
Deep fibromatosis Desmoid tumor MSK16	Clonal fibroblastic proliferation; sporadic CTNNB1 (β -catenin) or germline APC mutation (FAP/Gardner syndrome) → nuclear β -catenin accumulation. Pregnancy/estrogen and surgical trauma promote growth.	Extra-abdominal: shoulder, chest wall, thigh of teens/young adults. Abdominal wall (rectus): pregnant/postpartum women . Intra-abdominal (mesentery/pelvis): FAP/Gardner . Large, firm mass; locally aggressive, recurs, never metastasizes .	Gross: firm gray-white poorly circumscribed gritty mass. Micro: long sweeping fascicles of bland fibroblasts with collagen and slit-like vessels infiltrating skeletal muscle (trapped fibers). IHC: nuclear β-catenin+ .	Postpartum rectus mass or mesenteric mass in FAP = desmoid. Trap: "benign" (never metastasizes) yet infiltrative → needs wide excision with margin, high recurrence. Gardner: desmoid + osteomas + colon polyps.
LIPOMATOUS				
Lipoma Variants: angioliipoma, spindle cell/pleomorphic lipoma, chondroid lipoma MSK16	Most common soft tissue tumor; benign mature fat. Conventional lipomas often HMG A2 (12q14–15) rearrangement; spindle cell/pleomorphic lipoma 13q/RB1 loss. Multiple lipomas in Gardner syndrome, familial lipomatosis.	Adults 40–60; superficial subcutis of back, shoulder, neck, proximal extremities. Soft, mobile, painless, slow-growing. Cured by simple excision; no recurrence. Angioliipoma : young adults, forearm, multiple, painful .	Gross: thin-capsuled yellow lobulated fat. Micro: mature uniform adipocytes, no atypia. Angioliipoma: capillaries with fibrin thrombi ; spindle cell/pleomorphic: ropey collagen, floret giant cells (posterior neck, older men); chondroid: cartilage-like matrix.	Soft, mobile, superficial fatty mass = lipoma. Trap: deep or retroperitoneal fat = liposarcoma until proven otherwise ; liposarcoma never arises from a lipoma. Painful multiple forearm fatty nodules = angioliipoma.
Liposarcoma Well-differentiated/dedifferentiated, myxoid/round cell, pleomorphic MSK16	Most common adult sarcoma (with UPS); de novo, never from lipoma. WD/dedifferentiated: MDM2 (and CDK4) amplification, 12q13–15 ring/giant chromosomes. Myxoid/round cell: t(12;16) FUS-DDIT3 . Pleomorphic: complex karyotype.	Adults 40–60; deep thigh and retroperitoneum . WD: grade 1, recurs (esp. retroperitoneal), may dedifferentiate (high grade). Myxoid: 30–40s, thigh; more round cells = worse; mets to other soft tissue sites. Pleomorphic: older, high grade.	Key cell: lipoblast (vacuolated cytoplasm scalloping nucleus). WD: fat + fibrous bands + atypical hyperchromatic stromal cells; MDM2 FISH/IHC. Myxoid: gelatinous, chicken-wire capillaries ; round cell solid/opaque. Pleomorphic: bizarre lipoblasts.	Retroperitoneal fatty mass = liposarcoma. Lipoblast = diagnostic cell. Pair: t(12;16) FUS-DDIT3 = myxoid; MDM2 amplification = WD/dedifferentiated. Trap: vs intramuscular myxoma (no lipoblasts, scant vessels).
SKELETAL MUSCLE				

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
Rhabdomyosarcoma Embryonal (incl. botryoid), alveolar, pleomorphic MSK16	Malignant tumor with skeletal muscle differentiation. Alveolar: t(2;13) PAX3-FOXO1 or t(1;13) PAX7-FOXO1 (worse). Embryonal/pleomorphic: heterogeneous (embryonal 11p15 LOH). Li-Fraumeni, NF1, DICER1 predispose.	Most common childhood sarcoma. Embryonal (~50%, <10): head/neck (orbit), GU; botryoid in vagina/bladder of girls <5. Alveolar (10–25): deep extremities. Pleomorphic : men >60, thigh. High grade; can reach lymph nodes.	Micro: small round blue cells → strap cells; rhabdomyoblasts with eosinophilic cytoplasm ± cross-striations . Botryoid: grape-like, subepithelial cambium layer . Alveolar: discohesive cells around fibrous septa. IHC: desmin, myogenin, MyoD1 .	Grape-like vaginal mass in a young girl = botryoid embryonal RMS (cambium layer). Desmin + myogenin/MyoD1. Trap: arises where skeletal muscle is sparse (orbit, bladder); alveolar = older kids, t(2;13), worse prognosis.
VASCULAR				
Hemangioma Capillary (infantile/strawberry), cavernous MSK16	Benign endothelial neoplasm; most common tumor of infancy. Classified by vessel caliber. Multiple/visceral: von Hippel-Lindau (retina/cerebellum hemangioblastoma), Maffucci (with enchondromas). Large lesions trap platelets (Kasabach-Merritt).	Capillary/infantile : infants, head/neck skin, appears soon after birth, grows then regresses (by ~5–8 yr). Cavernous : larger, deeper, bluish, poorly circumscribed; does not regress ; liver = most common benign liver tumor.	Capillary: lobules of small capillaries, plump endothelium, inconspicuous lumens; GLUT1+ (infantile). Cavernous: large dilated blood-filled channels with flat endothelium, ± thrombi. Endothelial markers CD31/ERG.	Strawberry hemangioma regresses — observe (propranolol if complicated). Trap: don't biopsy hepatic cavernous hemangioma (bleeds). Large cavernous lesion + thrombocytopenia = consumptive coagulopathy (Kasabach-Merritt).
Angiosarcoma Stewart-Treves (lymphedema-associated) MSK16	Malignant endothelial (blood or lymphatic) tumor. Risk: chronic lymphedema (post-mastectomy, Stewart-Treves), prior radiation (breast), vinyl chloride , arsenic, Thorotrast (hepatic). Complex karyotype; post-radiation MYC amplification.	Elderly; scalp/face skin (bruise-like indurated patches); lymphedematous arm years after mastectomy; irradiated breast; deep soft tissue; liver. Red-brown nodules that ulcerate and bleed. Aggressive, high mortality; nodal and lung spread.	Micro: atypical endothelial cells lining irregular, anastomosing vascular channels dissecting collagen → solid sheets of pleomorphic cells in high grade. IHC: CD31, ERG , CD34, FLI1.	Post-mastectomy lymphedema + purple arm nodules = Stewart-Treves angiosarcoma. PVC worker + liver mass = hepatic angiosarcoma. Trap: Kaposi sarcoma is HHV-8, not angiosarcoma.
PERIPHERAL NERVE SHEATH				
Schwannoma Neurilemmoma; vestibular (acoustic) neuroma MSK16	Benign encapsulated tumor of pure Schwann cells (neural crest). NF2 (merlin/schwannomin) inactivation; NF2 syndrome : AD, chromosome 22, bilateral vestibular schwannomas , meningiomas, ependymomas, juvenile posterior subcapsular cataracts.	Adults; peripheral nerves (head/neck, flexor extremities) and CN VIII at cerebellopontine angle → hearing loss, tinnitus, imbalance. Grows eccentrically on the nerve → surgically separable, nerve spared. Malignant change very rare.	Gross: encapsulated, eccentric to nerve. Micro: cellular Antoni A (palisading spindle nuclei, Verocay bodies) alternating with loose myxoid Antoni B ; hyalinized vessels. IHC: diffuse strong S-100 , SOX10.	Verocay bodies + Antoni A/B + S-100 = schwannoma. Bilateral acoustic schwannomas = NF2 ("NF2 = 2 ears, chr 22"). Trap: schwannoma pushes nerve aside; neurofibroma grows through it.
Neurofibroma Cutaneous, intraneural, diffuse, plexiform MSK16	Benign unencapsulated mixed nerve-sheath tumor (Schwann cells, fibroblasts, perineurial cells, entrapped axons). NF1 : AD, chr 17, neurofibromin (RAS-GAP) loss → RAS stuck on → MAPK proliferation.	Cutaneous: soft solitary skin nodules, often sporadic. Plexiform : childhood, large nerves, disfiguring; essentially diagnostic of NF1 ; ~10% lifetime MPNST risk. NF1: café-au-lait, axillary/inguinal freckling, Lisch nodules, optic glioma, bone dysplasia.	Gross: plexiform = " bag of worms " of thickened fascicles; intraneural = fusiform nerve enlargement. Micro: wavy bland spindle cells, " shredded carrot " collagen in myxoid stroma, mast cells, axons within tumor. S-100 (patchy).	Plexiform neurofibroma = NF1 and MPNST precursor. Lisch nodules specific for NF1; café-au-lait alone is not. Trap: rapid growth/pain in plexiform lesion → suspect MPNST transformation.
Malignant peripheral nerve sheath tumor MPNST MSK16	Sarcoma of nerve sheath origin; ~50% arise in NF1 (from plexiform neurofibroma, ~10% lifetime risk), others sporadic or post-radiation . NF1 + CDKN2A loss + PRC2 (SUZ12/EED) loss → H3K27me3 loss.	Adults 20–50 (younger in NF1); large proximal nerves of extremities, trunk, sciatic nerve, brachial plexus. Rapidly enlarging painful mass ± neurologic deficit. High grade, local recurrence, hematogenous lung mets; poor prognosis.	Gross: fusiform mass along a major nerve, necrosis. Micro: fascicles of hyperchromatic spindle cells, alternating hyper-/hypocellular ("marbled") areas, perivascular accentuation, necrosis, mitoses. IHC: S-100/SOX10 weak/focal , loss of H3K27me3.	NF1 patient + plexiform neurofibroma that suddenly grows or hurts = MPNST. Trap: S-100 only patchy (vs diffuse strong in schwannoma); sarcomas spread to lung, not lymph nodes.
UNCERTAIN / OTHER				

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
Synovial sarcoma MSK16	Sarcoma of uncertain histogenesis — not from synovium. t(X;18) SS18-SSX (SSX1/SSX2) in >90%; simple translocation karyotype. Epithelial differentiation despite mesenchymal origin.	Young adults 15–40; deep soft tissue near large joints , especially around the knee (thigh, leg). Slowly enlarging, often painful mass. High grade by convention; lung mets and one of the sarcomas spreading to lymph nodes .	X-ray: soft tissue mass with calcifications . Micro: biphasic (epithelial glands + uniform spindle cells) or monophasic spindle. IHC: keratin/EMA+ , TLE1+ (and SS18-SSX fusion-specific antibody).	"X marks the knee of an 18-year-old" = t(X;18) SS18-SSX. Biphasic + keratin+ spindle-cell sarcoma near knee. Trap: name misleads — does not arise from synovium; usually extra-articular.
Intramuscular myxoma MSK18	Benign tumor of uncertain histogenesis within skeletal muscle; activating GNAS mutations. Multiple intramuscular myxomas + polyostotic fibrous dysplasia = Mazabraud syndrome .	Adults 40–70, women; large muscles of thigh, shoulder, buttock. Painless, slowly growing, deep, mobile mass. Benign; does not recur after complete excision ; no metastasis.	Gross: white, glistening, gelatinous nodules within muscle . Micro: very few bland stellate/spindle cells in abundant pale myxoid matrix with scant vessels ; no lipoblasts, no atypia.	Pale gelatinous hypocellular, hypovascular intramuscular mass = myxoma. Trap: vs myxoid liposarcoma (lipoblasts + chicken-wire capillaries, t(12;16)). Myxomas + fibrous dysplasia = Mazabraud (GNAS).
Undifferentiated pleomorphic sarcoma UPS; formerly malignant fibrous histiocytoma (MFH) MSK16	Diagnosis of exclusion: high-grade pleomorphic sarcoma with no identifiable line of differentiation. Complex karyotype , genomic instability; can follow prior radiation . Many older MFH cases reclassified by IHC/molecular work-up.	One of the most common adult sarcomas (with liposarcoma); older adults 50–70; deep thigh , extremities, retroperitoneum. Large, painless, enlarging deep mass. High grade, recurs, hematogenous lung mets; aggressive.	Gross: large fleshy gray-white mass with hemorrhage and necrosis. Micro: sheets of bizarre pleomorphic spindle and giant cells, storiform pattern, atypical mitoses, necrosis. IHC: negative for lineage markers (desmin, S-100, keratin, MDM2).	Older adult + large deep thigh mass + highly pleomorphic cells + lineage IHC all negative = UPS. Trap: must exclude dedifferentiated liposarcoma (MDM2), pleomorphic RMS, sarcomatoid carcinoma, melanoma first.

Peripheral Nerve & Muscle

Neuropathies, motor neuron, NMJ and myopathies · 28 shown

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
PATTERNS OF NERVE INJURY				
Axonal degeneration & regeneration Wallerian degeneration MSK19	Primary injury to axon/neuron (~80% of neuropathies); distal axon + its myelin break down after transection or metabolic/toxic/ischemic injury. Causes: diabetes (mostly), toxins, vasculitis, B-vitamin deficiency, alcohol, trauma.	Distal, length-dependent sensorimotor loss (stocking-glove); weakness with denervation atrophy of muscle. Recovery slow via axonal sprouting; only peripheral axons regenerate — CNS neurons cannot.	NCS: ↓ amplitude , near-normal conduction velocity. Biopsy: myelin ovoids , macrophages clearing debris, fiber loss, regenerating clusters; EM swollen axons with degenerating organelles, dense bodies. Muscle: denervation atrophy.	Myelin ovoids; ↓ amplitude = axonal. Most common pattern (~80%). Trap: axonal injury → denervation atrophy of muscle, demyelination spares muscle early. Vasculitic infarcts can cause both patterns.
Segmental demyelination Onion bulbs MSK19	Primary Schwann cell/myelin injury; macrophages strip myelin from internodes while axon survives. Repeated demyelination/remyelination → concentric Schwann cell layers. Causes: GBS, CIDP, CMT1, lepromatous leprosy, diphtheria.	Weakness and sensory loss, often proximal + distal, with early areflexia; little muscle atrophy early. Acute (GBS) or chronic/relapsing (CIDP, CMT1).	NCS: ↓ conduction velocity , conduction block. Biopsy: bare (denuded) axons, thin short remyelinated internodes, variable myelin thickness; repeated cycles → onion bulbs .	Onion bulbs = 'Over and Over' demyelination/remyelination → CIDP, CMT1. ↓ velocity = demyelinating vs ↓ amplitude = axonal. Axon preserved → good recovery potential.
NEUROPATHIES				
Guillain-Barré syndrome AIDP MSK19	Acute immune-mediated demyelinating radiculoneuropathy; ~2/3 follow flu-like/GI illness — Campylobacter jejuni , CMV, EBV, Mycoplasma. Molecular mimicry: antibodies + lymphocytes, not organism, damage roots and nerves.	Days–weeks: distal weakness rapidly ascending ± paresthesias, areflexia; autonomic instability, facial diplegia possible. Life-threatening respiratory failure → monitor vital capacity. Most recover.	CSF albuminocytologic dissociation (↑ protein, normal cells). NCS: slowed velocity, conduction block. Pathology: inflammation + segmental demyelination of roots/nerves. Rarely biopsied; diagnosis clinical + CSF + NCS.	Ascending paralysis after Campylobacter diarrhea, areflexia. Treat IVIG or plasmapheresis — NOT steroids. Trap: damage is immune (molecular mimicry), not infection. Watch vital capacity.
CIDP Chronic inflammatory demyelinating polyradiculoneuropathy MSK19	Chronic immune-mediated demyelination of roots and nerves; no infectious trigger identified. Repeated demyelination/remyelination cycles → Schwann cell proliferation, hypertrophic nerves.	Subacute/chronic, progressive or relapsing (>8 weeks); often symmetric sensory + motor weakness, proximal and distal, hyporeflexia.	NCS: slowed velocity, conduction block. CSF ↑ protein. Sural biopsy: demyelination/remyelination, onion bulbs , variable myelin thickness, short internodes.	'Chronic GBS' (>8 wk) with onion bulbs. Unlike GBS, responds to steroids (also IVIG, plasmapheresis). Relapsing symmetric sensorimotor neuropathy + onion bulbs = CIDP.
Diabetic neuropathy MSK19	Most common neuropathy; affects most diabetics. Microvascular ischemia (diabetic vasculopathy) + nonenzymatic glycation/ AGEs + polyol (sorbitol) pathway. Predominantly axonal.	Years: symmetric distal sensory/sensorimotor stocking-glove neuropathy, neuropathic ulcers, Charcot joints; autonomic (gastroparesis, orthostasis, ED); focal/multifocal mononeuropathy (CN III with pupil sparing).	NCS: mainly axonal (↓ amplitude). Biopsy: loss of small and large myelinated fibers, thinly myelinated regenerating fibers, thickened hyalinized endoneurial arterioles .	Stocking-glove + thick hyalinized endoneurial arterioles. Diabetic CN III palsy spares pupil (ischemia hits central fibers; parasympathetics peripheral). Loss of pain → ulcers, Charcot joint.
Herpes zoster Shingles; postherpetic neuralgia MSK19	VZV latent in sensory dorsal root and cranial nerve ganglia after chickenpox; declining cell-mediated immunity (age, immunosuppression) → reactivation, travels down sensory nerve.	Older/immunosuppressed: burning pain then painful vesicular rash in one dermatome , not crossing midline. Complications: postherpetic neuralgia (pain >3 mo), zoster ophthalmicus (V1), Ramsay Hunt.	Ganglia: neuronal destruction and loss with abundant mononuclear inflammation; intranuclear inclusions usually not found. Tzanck (skin): multinucleated giant cells; PCR confirms.	Dermatomal vesicles after prodromal burning pain; latent in sensory ganglia. Trap: ganglion inclusions usually absent. Prevent with recombinant zoster vaccine; antivirals (acyclovir) ↓ duration.

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
Leprosy neuropathy Hansen disease MSK19	Mycobacterium leprae invades Schwann cells ; prefers cool peripheral nerves/skin. Tuberculoid (Th1): granulomas destroy nerves. Lepromatous (Th2): bacilli proliferate, segmental demyelination.	Immigrant/endemic exposure; anesthetic hypopigmented skin patches, thickened palpable nerves (ulnar, peroneal), injury and loss of numb digits. Tuberculoid localized; lepromatous diffuse, leonine facies.	Nerve biopsy (indicated if suspected): tuberculoid = granulomas, few bacilli; lepromatous = many acid-fast bacilli in Schwann cells and macrophages (foam cells), segmental demyelination.	Numb skin patches + thickened nerves = leprosy; grows in cool sites (armadillo reservoir). Tuberculoid few AFB vs lepromatous many AFB. Treat dapsone + rifampin (± clofazimine).
Charcot-Marie-Tooth disease Hereditary motor and sensory neuropathy MSK19	Hereditary neuropathy; most common CMT1A = PMP22 duplication (AD) → demyelinating; CMT2 axonal (e.g., MFN2); X-linked form connexin-32 (GJB1). Defective myelin/axon structural proteins.	Childhood–adolescence, slowly progressive distal weakness/atrophy and sensory loss; pes cavus , hammer toes, foot drop, 'stork legs' (inverted champagne bottle); ↓ reflexes.	CMT1: markedly, uniformly ↓ conduction velocity. Biopsy: demyelination/remyelination with onion bulbs , hypertrophic nerves. Now diagnosed by genetic testing rather than biopsy.	Pes cavus + hammer toes + stork legs + onion bulbs = CMT1A (PMP22 duplication). Trap: HNPP = PMP22 deletion (recurrent pressure palsies). Onion bulbs: CMT1 and CIDP.
Vasculitic neuropathy MSK19	Necrotizing vasculitis of epineurial arteries → nerve infarcts; causes axonal degeneration ± demyelination. Polyarteritis nodosa, microscopic polyangiitis, EGPA, cryoglobulinemia, RA, SLE; also isolated nerve vasculitis.	Painful, asymmetric/multifocal neuropathy — mononeuritis multiplex (sequential wrist drop, foot drop); systemic vasculitis features (fever, renal, skin).	NCS: multifocal axonal (↓ amplitude). Sural nerve + muscle biopsy: transmural inflammation, fibrinoid necrosis of epineurial vessels, asymmetric fiber loss between fascicles. ANCA, cryoglobulins, hepatitis B/C.	Key indication for sural nerve biopsy (with vasculitis, amyloid, sarcoid, leprosy). Mononeuritis multiplex → think PAN, EGPA, diabetes. Result changes treatment (immunosuppression).
Compression/entrapment neuropathy Carpal tunnel; Morton neuroma STEP 1	Chronic focal compression → ischemia, demyelination then axonal loss. Carpal tunnel: median nerve under flexor retinaculum (pregnancy, RA, hypothyroid, DM, amyloid, acromegaly, dialysis). Morton: perineural fibrosis of plantar interdigital nerve.	Carpal tunnel: nocturnal hand paresthesias lateral 3½ digits, thenar atrophy, Tinel/Phalen + , palm sensation spared. Morton: women in tight/high-heeled shoes, burning pain between 3rd–4th metatarsals .	NCS: focal slowing/conduction block across site, later ↓ amplitude. Morton neuroma: not a true neoplasm — perineural fibrosis, degenerated nerve fibers, thickened vessels; clicking on squeeze (Mulder sign).	Bilateral carpal tunnel → think hypothyroid, pregnancy, dialysis-related β2-microglobulin amyloid, acromegaly. Palmar cutaneous branch spares palm. Morton neuroma = reactive fibrosis, not tumor.
MOTOR NEURON				
Amyotrophic lateral sclerosis Lou Gehrig disease MSK19	Degeneration of upper + lower motor neurons (corticospinal tracts, anterior horn, brainstem motor nuclei). Mostly sporadic; familial: SOD1 , C9orf72 repeat, TDP-43.	Middle-aged adults: asymmetric limb weakness; LMN (atrophy, fasciculations) + UMN (spasticity, hyperreflexia, Babinski); bulbar dysarthria/dysphagia. Sensation, eye movements, bladder spared. Death ~3–5 yr from respiratory failure.	EMG: widespread denervation (fibrillations, fasciculations). Muscle biopsy: neurogenic — angular atrophic fibers, fiber-type grouping, grouped atrophy ; loss of checkerboard. CK normal/mildly ↑; genetic testing confirms.	Mixed UMN + LMN signs with no sensory loss. Biopsy: checkerboard lost → fiber-type grouping, grouped atrophy. Riluzole (↓ glutamate) modestly prolongs survival; SOD1 familial.
Spinal muscular atrophy Werdnig-Hoffmann (type I) MSK19	Loss of anterior horn motor neurons from SMN1 mutation/deletion, chromosome 5q , autosomal recessive; SMN2 copy number modifies severity.	Type I: floppy baby — severe hypotonia, limb/axial weakness, poor head control, weak cry and suck, tongue fasciculations , respiratory failure. LMN only; milder types later onset.	Muscle biopsy: large groups of atrophic fibers with a few scattered hypertrophic (still-innervated) fibers — neurogenic grouped atrophy. EMG denervation. Diagnosis now by genetics, not biopsy.	Floppy infant + tongue fasciculations = SMA (LMN, anterior horn), 'infantile polio.' Treat nusinersen (SMN2 splicing antisense) or gene therapy. Trap: hypertrophic fibers retain innervation.
Denervation atrophy pattern Neurogenic atrophy MSK19	Motor neuron/axon lost → orphaned fibers atrophy. Surviving neighboring axons sprout, reinnervate and convert fibers to their own type. If that enlarged motor unit dies → whole group atrophies.	Stage 1 normal → Stage 2 scattered small angular atrophic fibers → Stage 3 fiber-type grouping (checkerboard lost) → Stage 4 grouped atrophy of same-type fibers. Seen in ALS, SMA, neuropathies.	Esterase: angular atrophic fibers stain dark. ATPase (pH 9.4/4.3): loss of type 1/type 2 checkerboard → large same-type groups. Little necrosis/regeneration; internal nuclei, fibrosis only late.	Neurogenic = angular atrophy, fiber-type grouping; myopathic = rounded fibers, necrosis, checkerboard kept. The nerve sets the fiber type. Only peripheral axons reinnervate; CNS neurons cannot.
NEUROMUSCULAR JUNCTION				

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
Myasthenia gravis MSK19	Autoantibodies against postsynaptic nicotinic ACh receptors (type II hypersensitivity; complement, receptor internalization) → ↓ endplate potential. Anti-MuSK in some seronegative. Thymic hyperplasia/thymoma association.	Young women/older men: fluctuating fatigable weakness worse with use and evening; ptosis, diplopia, bulbar, proximal limbs. Myasthenic crisis → respiratory failure. Reflexes, sensation normal.	Diagnosed by antibodies + EMG , not biopsy: anti-AChR (± MuSK), decremental response on repetitive stimulation, ice-pack test, single-fiber EMG jitter. Chest CT for thymoma.	Muscle biopsy not indicated. Fatigable ptosis/diplopia, improves with rest/AChE inhibitors (pyridostigmine). Thymoma → thymectomy. Trap: aminoglycosides, β-blockers, Mg worsen it. Decrement vs LEMS increment.
Lambert-Eaton myasthenic syndrome LEMS MSK19	Autoantibodies against presynaptic voltage-gated Ca2+ channels (P/Q-type) → ↓ ACh release. Paraneoplastic — small cell lung carcinoma — or autoimmune.	Proximal limb weakness (legs) that improves with repeated use ; hypo/areflexia; autonomic symptoms (dry mouth, constipation, ED); ocular involvement minimal.	Diagnosed by antibodies + EMG , not biopsy: anti-VGCC; low baseline CMAP with incremental response on high-frequency repetitive stimulation or after brief exercise. Search for SCLC.	Muscle biopsy not indicated. Weakness improves with use + ↓ reflexes + autonomic = LEMS → CT chest (SCLC). Treat tumor, 3,4-diaminopyridine (blocks K+ channels). Increment vs MG decrement.
Botulism STEP 1	Clostridium botulinum preformed toxin (adults: canned food) or spores (infants: honey , soil). Toxin cleaves SNARE proteins → blocks presynaptic ACh release at NMJ and autonomic synapses.	Descending symmetric flaccid paralysis: diplopia, ptosis, dysarthria, dysphagia ('4 Ds'), fixed dilated pupils, then limb and respiratory weakness. Infant: floppy baby, constipation, weak cry/suck. Sensation, mentation intact.	Clinical diagnosis; toxin in serum/stool/food (mouse bioassay), C. botulinum in infant stool. EMG: small CMAP, incremental response at high-frequency stimulation. CSF normal.	Descending (vs GBS ascending) paralysis with pupil involvement; infant + honey. Treat antitoxin (BabyBIG infants). Trap: tetanus blocks glycine/GABA release from Renshaw cells → spastic, botulism → flaccid.
DYSTROPHIES				
Duchenne muscular dystrophy DMD MSK19	X-linked (Xp21) dystrophin frameshift /deletion → no functional dystrophin; dystrophin links cytoskeleton to ECM via dystroglycan complex. Largest gene (~2.5 Mb, 79 exons); ~1/3 new mutations.	Boys, weakness by ~age 5: proximal/pelvic, Gowers maneuver , waddling gait, calf pseudohypertrophy (fibrofatty). Wheelchair by 10–12; death early 20s from dilated cardiomyopathy or respiratory failure.	Markedly ↑ CK . Dystrophic biopsy: fiber size variation, necrotic/regenerating fibers, internal nuclei, endomysial fibrosis, fat, no inflammation. Absent dystrophin on IHC; no band on Western; genetics.	Duchenne = Deleted (frameshift) dystrophin → absent protein. Calf pseudohypertrophy = fat/fibrosis, not muscle. Absent dystrophin IHC confirms. Death from cardiomyopathy/respiratory failure.
Becker muscular dystrophy BMD MSK19	X-linked dystrophin in-frame deletion → truncated, partially functional dystrophin. Less common than Duchenne; same Xp21 gene and dystrophic process.	Boys/young men, later onset (adolescence–adulthood) and milder course than DMD; proximal weakness, calf pseudohypertrophy, cramps; dilated cardiomyopathy may dominate. Survive into adulthood.	↑ CK. Dystrophic biopsy (milder). Dystrophin weak/patchy on IHC; Western blot band of abnormal size or amount . Gene sequencing confirms.	Becker = Better (in-frame). Trap: Western reduced/abnormal-size band (Becker) vs no band (Duchenne). Reading frame, not deletion size, predicts severity.
Limb-girdle muscular dystrophy LGMD MSK19	Autosomal recessive or dominant defects in dystrophin-associated/membrane proteins: sarcoglycans , caveolin (also dysferlin, calpain-3). Related: α2-laminin (merosin) → congenital MD.	Both sexes, childhood–adulthood: proximal hip and shoulder girdle weakness; face usually spared; variable progression, cardiac involvement in some.	↑ CK. Dystrophic biopsy; missing sarcoglycan (or caveolin) on IHC with normal dystrophin. Most now diagnosed by clinical exam + genetic panels.	Duchenne-like girdle weakness in a girl or with autosomal inheritance → LGMD (sarcoglycan). Other dystrophies: Emery-Dreifuss, facioscapulohumeral, oculopharyngeal.
Myotonic dystrophy DM1 (Steinert); DM2 MSK19	AD repeat expansion. DM1: CTG trinucleotide in DMPK , chromosome 19 → toxic RNA; anticipation (esp. maternal). DM2: CCTG tetranucleotide in ZNF9 (CNBP), chromosome 3, milder.	Adults: myotonia (can't release handshake), distal and facial weakness (hatchet face), cataracts, frontal balding, testicular atrophy, cardiac arrhythmias/conduction block, insulin resistance. Congenital DM1: hypotonia, learning disability.	EMG: myotonic discharges ('dive-bomber'). Biopsy: many internal (central) nuclei , ring fibers, type 1 fiber atrophy. Diagnosis by genetic repeat testing.	Can't release grip + cataracts + balding + testicular atrophy = CTG/DMPK. Most common adult dystrophy. Anticipation; congenital form from affected mother. Death from arrhythmia — ECG monitoring.
CONGENITAL / METABOLIC				

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
Nemaline myopathy Rod myopathy MSK19	Congenital myopathy; sarcomeric thin-filament genes: ACTA1 (skeletal actin, usually dominant/de novo), nebulin (NEB) , recessive); rarely tropomyosins, troponin T, cofilin.	Floppy infant : hypotonia and weakness of face, neck and respiratory muscles; feeding difficulty, respiratory failure; non-progressive or slowly progressive.	Gomori trichrome: rod-shaped purple cytoplasmic inclusions. EM: rods of Z-band density. CK normal/mildly ↑; genetic testing confirms.	Floppy baby + purple rods on trichrome (Z-band material). DDx floppy baby: SMA (tongue fasciculations, grouped atrophy), infant botulism, Pompe, congenital DM1.
Mitochondrial myopathy MELAS, MERRF, Kearns-Sayre, Leigh MSK19	mtDNA mutations → oxidative phosphorylation defects; maternal inheritance + heteroplasmy (variable severity). KSS usually sporadic mtDNA deletion; Leigh often nuclear genes.	MELAS : stroke-like episodes, lactic acidosis, seizures. MERRF : myoclonus, seizures, ataxia. KSS (onset under 20 yr): PEO, pigmentary retinopathy, heart block. Leigh: infant ataxia, seizures. Exercise intolerance, hearing loss, short stature.	Gomori trichrome: ragged red fibers (subsarcolemmal mitochondrial clumps). EM: 'parking-lot' paracrystalline inclusions. ↑ serum lactate (lactic acidosis).	Ragged red fibers + maternal inheritance + lactic acidosis. All offspring of affected mother at risk; none from father. Heteroplasmy → variable expression within family. KSS = heart block.
Malignant hyperthermia STEP 1	AD RYR1 mutation (sometimes CACNA1S) in sarcoplasmic reticulum → uncontrolled Ca ²⁺ release with inhaled anesthetics (halothane, sevoflurane) or succinylcholine → sustained contraction, hypermetabolism.	During/after anesthesia: ↑ end-tidal CO ₂ (earliest), masseter rigidity, generalized rigidity, tachycardia, hyperthermia (late), rhabdomyolysis, hyperkalemia, arrhythmias. Associated with central core disease.	Mixed respiratory + metabolic (lactic) acidosis, hyperkalemia, markedly ↑ CK, myoglobinuria. Susceptibility: caffeine-halothane contracture test on muscle biopsy; RYR1 genetic testing.	Anesthesia + rigidity + ↑ CO ₂ + fever → stop trigger, give dantrolene (RyR antagonist). Trap: NMS (antipsychotics) also treated with dantrolene; both cause rhabdomyolysis.
INFLAMMATORY				
Polymyositis MSK19	T-cell mediated: CD8+ cytotoxic T cells attack muscle fibers expressing MHC-I. Associated antibodies: anti-Jo-1 (antisynthetase), ANA.	Adults (>18), F:M 2:1: subacute symmetric proximal weakness (rising from chair, climbing stairs, lifting arms); no rash. May have interstitial lung disease, dysphagia.	CK ↑ (up to 50×). EMG myopathic. Biopsy: endomysial CD8+ T cells surrounding and invading non-necrotic fibers ; necrotic/regenerating fibers scattered throughout fascicle; MHC-I upregulation.	Polymyositis = Penetrating CD8 T cells inside the fascicle (endomysial). No rash. Anti-Jo-1 → ILD, mechanic's hands. Treat steroids. Trap: vs dermatomyositis perifascicular.
Dermatomyositis MSK19	Antibody/ complement-mediated microvasculopathy (capillary loss, ischemia at fascicle edge); CD4+ T cells and B cells. Paraneoplastic in adults (ovarian, lung, GI).	Children (5–15) and adults (45–65), F:M 2:1: symmetric proximal weakness + rash — heliotrope eyelids, Gotttron papules , shawl sign, nailfold changes. Calcinosis in children. Malignancy screening in adults.	CK normal to ↑ (50×). Biopsy: perifascicular atrophy , perivascular/perimysial B cells + CD4+ T cells. Anti-Mi-2 specific; anti-Jo-1 (antisynthetase), ANA.	Damage at the eDge = perifascicular atrophy. Trap: atrophy is perifascicular, inflammation is perivascular. Heliotrope + Gotttron → screen for cancer. Anti-Mi-2 specific.
Inclusion body myositis IBM MSK19	Degenerative + inflammatory myopathy of older adults; abnormal protein aggregates (tubulofilamentous inclusions, amyloid, TDP-43) with endomysial T-cell inflammation.	Usually men >50 ; slowly progressive, often asymmetric distal + proximal weakness: finger flexors , wrist flexors, ankle dorsiflexors (foot drop), quadriceps ; dysphagia. Most common myopathy >50.	CK normal to mildly ↑. Biopsy: rimmed vacuoles ; tubulofilamentous inclusions in cytoplasm and nucleus (EM); endomysial T cells invading fibers; ragged red/COX-negative fibers possible.	'Can't grip a cup, can't climb up.' Old man + finger flexor/quadriceps weakness + rimmed vacuoles. Steroid-resistant — the polymyositis that fails steroids is IBM.
TOXIC / ENDOCRINE				
Statin myopathy MSK19	HMG-CoA reductase inhibitors → direct myotoxicity (↓ CoQ10/isoprenoids); ↑ risk with high dose, fibrates, CYP3A4 inhibitors. Rare: anti-HMGCR immune-mediated necrotizing myopathy.	Symmetric proximal myalgia, cramps and weakness; mild–moderate CK elevation; occasionally rhabdomyolysis (myoglobinuria, AKI). Resolves after stopping statin (except anti-HMGCR).	CK mild–moderately ↑ (marked in rhabdo). Biopsy (if persistent): necrotic and regenerating fibers with little inflammation; anti-HMGCR antibodies in autoimmune form.	Management: stop statin, switch lipid-lowering agent. Trap: weakness persisting after stopping + high CK → anti-HMGCR necrotizing autoimmune myopathy (needs immunotherapy). Gemfibrozil ↑ risk.
Glucocorticoid myopathy Steroid myopathy STEP 1	Chronic exogenous glucocorticoids or Cushing syndrome → protein catabolism, ↓ protein synthesis, preferentially in type II (fast, glycolytic) fibers.	Insidious painless proximal weakness, legs > arms (rising from chair, stairs), with cushingoid features; no myalgia. Improves over weeks–months after steroid taper.	CK normal ; EMG normal or mildly myopathic. Biopsy: type II fiber atrophy (ATPase), no necrosis or inflammation.	Proximal weakness + normal CK in a steroid-treated patient = steroid myopathy. Trap: myositis on steroids with worsening weakness — normal CK favors steroid myopathy over flare. Type II atrophy also disuse.

Rheumatology

Joint disease: degenerative, autoimmune, crystal, infectious, seronegative · 11 shown

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
DEGENERATIVE / AUTOIMMUNE				
Osteoarthritis Degenerative joint disease MSK11	Most common joint disease: chondrocyte injury → MMPs degrade type II collagen and proteoglycans faster than repair → fibrillation → cartilage loss. Minimal, secondary inflammation. Primary (aging) or secondary: injury, obesity, AVN, diabetes, hemochromatosis.	Older, obese, or prior injury; asymmetric , weight-bearing joints (knees, hips), spine, DIP , PIP, 1st CMC. Pain worse with use, better with rest; morning stiffness under 30 min . No systemic disease.	X-ray: asymmetric joint-space narrowing, osteophytes , subchondral sclerosis, subchondral cysts. Gross: fibrillation, eburnation , joint mice. Synovial WBC under 2,000. No autoantibodies.	Heberden (DIP) and Bouchard (PIP) nodes = bony osteophytes. Degenerative, not autoimmune: no RF/anti-CCP. Trap: DIP involvement favors OA (or psoriatic), RA spares DIP.
Rheumatoid arthritis MSK11	HLA-DR4 + smoking → citrullinated self proteins → CD4 T cells + anti-CCP (ACPA) and RF (IgM vs Fc of IgG). TNF , IL-1, IL-6 → synovial hyperplasia → pannus erodes cartilage/bone (RANKL/osteoclasts). Women > men.	Symmetric small joints: MCP, PIP, wrist ; spares DIP. Morning stiffness >1 h , improves with activity. Late: ulnar deviation, swan-neck, boutonnière; C1–C2 subluxation. Nodules, lung, heart, vessel, eye disease; Felty, Caplan, amyloidosis.	X-ray: symmetric narrowing, periarticular osteopenia , marginal erosions. Synovium: villous hyperplasia, lymphoplasmacytic follicles, rice bodies , pannus. Nodule: central fibrinoid necrosis + palisading macrophages . Anemia of chronic disease.	Anti-CCP most specific ; RF sensitive, not specific. Felty = RA + splenomegaly + neutropenia. Check C1–C2 before intubation. Rheumatoid nodule = soft tissue, not bony. Tx: methotrexate, anti-TNF.
Juvenile idiopathic arthritis MSK11	Chronic idiopathic arthritis with onset before age 16 , lasting ≥6 weeks. Autoimmune (HLA-associated); subtypes oligoarticular (most common), polyarticular, systemic (Still disease—IL-1/IL-6 driven).	More large-joint involvement and systemic disease than adult RA. Systemic: quotidian spiking fever, salmon-colored evanescent rash, HSM, serositis. Complications: uveitis (oligoarticular, ANA+), pericarditis, myocarditis, glomerular disease, pulmonary fibrosis, growth disturbance.	Often RF-negative ; ANA+ in oligoarticular (uveitis risk). Systemic: ↑ ESR/CRP, ferritin, leukocytosis. Synovium resembles adult RA. X-ray: soft-tissue swelling, periarticular osteopenia, growth abnormalities.	Onset before 16, large joints, RF-negative, systemic features. ANA+ oligoarticular JIA → chronic anterior uveitis (often asymptomatic) → routine slit-lamp exams. Systemic JIA risk: macrophage activation syndrome.
CRYSTAL				
Gout MSK11	Hyperuricemia (necessary, not sufficient) → monosodium urate precipitates. Underexcretion (most; renal disease, thiazide/loop diuretics , low-dose aspirin, alcohol) or overproduction (HGPRT deficiency—Lesch-Nyhan, tumor lysis, myeloproliferative). Crystals → NLRP3 inflammasome → IL-1 , neutrophils.	Middle-aged men; sudden, excruciating red, hot joint— 1st MTP (podagra) —often at night after alcohol, big meal, diuretic. Stages: asymptomatic hyperuricemia → acute → intercritical → chronic tophaceous. Urate stones, nephropathy.	Aspirate: needle-shaped, negatively birefringent crystals (yellow parallel) inside neutrophils. Tophus: pale crystal aggregates (dissolve in formalin) + foreign-body giant cells. X-ray: punched-out erosions with overhanging edges .	Gout = Needles, Negative ; YUP (Yellow = Urate, Parallel). Tophi: ear helix, olecranon, Achilles. Acute: NSAIDs, colchicine, steroids; start allopurinol/febuxostat only after flare . Serum urate may be normal during attack.
Pseudogout CPPD deposition disease MSK11	Calcium pyrophosphate dihydrate deposition; pathogenesis unclear, perhaps cartilage degradation. Associations: hyperparathyroidism, hemochromatosis, hypothyroidism, hypomagnesemia ; aging.	Elderly ; mono/oligoarticular: knee (most), wrist, elbow, shoulder. Acute hot joint mimicking gout or septic arthritis; can cause significant joint damage; often coexists with OA.	Aspirate: rhomboid, weakly positively birefringent crystals (blue parallel). X-ray: chondrocalcinosis (linear cartilage calcification). Histo: basophilic (purple) crystal deposits in articular cartilage.	Pseudogout = Positive, Pyrophosphate, Parallelogram (rhomboid), blue parallel. Elderly knee + chondrocalcinosis → check Ca ²⁺ /PTH, iron studies, Mg ²⁺ . Trap: gout is needle/negative/1st MTP.
INFECTIOUS				

DISEASE	PATHOGENESIS · GENETICS	WHO · WHERE · CLINICAL	IMAGING · LABS · PATHOLOGY	HIGH YIELD
Septic arthritis Incl. gonococcal arthritis MSK11	Hematogenous seeding (or direct inoculation, prosthesis) → neutrophil-mediated rapid cartilage destruction. S. aureus most common overall; N. gonorrhoeae in young sexually active adults. Risk: RA, prosthetic joint, IVDU, immunosuppression, sickle cell.	Fever + single hot, swollen, painful joint (knee most) with ↓ ROM. Gonococcal : young sexually active; triad of migratory polyarthralgia, tenosynovitis (hand), vesiculopustular skin lesions, or monoarthritis. Complication: joint destruction, sepsis.	Synovial fluid: WBC >50,000, neutrophil-predominant , ↓ glucose; Gram stain/culture positive (<i>S. aureus</i> : GPC in clusters). Gonococcal: synovial culture often negative → culture/NAAT urethra, cervix, pharynx, rectum. Blood cultures.	Medical emergency → aspirate before antibiotics , drain. Monoarticular hot joint in young sexually active = gonococcus. Crystals do not exclude infection. Treat Chlamydia empirically with gonococcal.
Lyme arthritis STEP 1	Borrelia burgdorferi (spirochete) via Ixodes deer tick; Northeast/upper Midwest US. Stage 3 (late disseminated) → immune-mediated synovitis months after infection; antibiotic-refractory form linked to HLA-DR4.	Hiker in Northeast: early erythema migrans (bull's-eye), then facial nerve (Bell) palsy, AV block, migratory arthralgia; late: intermittent monoarticular/oligoarticular arthritis of the knee , encephalopathy.	Synovial fluid inflammatory (WBC ~10,000–25,000), culture negative; ELISA → Western blot serology; PCR of synovial fluid. Synovium: villous hyperplasia, lymphoplasmacytic infiltrate, obliterative arteritis.	Large-joint (knee) monoarthritis + tick exposure = Lyme. Bilateral Bell palsy buzzword. Tx: doxycycline (amoxicillin in children/pregnancy); ceftriaxone for CNS/high-grade AV block.
SERONEGATIVE SPONDYLOARTHROPATHIES (STEP 1)				
Ankylosing spondylitis STEP 1	HLA-B27 -associated seronegative (RF–) spondyloarthropathy; IL-17/IL-23 and TNF-driven enthesitis → reactive new bone formation → ossification of ligaments and fusion. Young men > women.	Young man with chronic inflammatory low back pain : morning stiffness, better with exercise, worse with rest; sacroiliitis , ↓ chest expansion, kyphosis, Achilles enthesitis. Extra-articular: anterior uveitis , aortic regurgitation (aortitis), restrictive lung disease.	X-ray/MRI: bilateral sacroiliitis, bamboo spine (syndesmophytes, vertebral fusion), squaring of vertebrae. RF and ANA negative; HLA-B27+; ↑ ESR/CRP.	PAIR seronegatives (Psoriatic, Ankylosing, IBD, Reactive): HLA-B27, RF–. Bamboo spine + uveitis + aortic regurgitation. Fused spine → fracture with minor trauma. Tx: NSAIDs, anti-TNF, IL-17 inhibitors.
Reactive arthritis STEP 1	HLA-B27 post-infectious sterile arthritis 1–4 wk after GI (Shigella, Salmonella, Yersinia , Campylobacter) or GU (Chlamydia) infection; molecular mimicry. Young men.	" Can't see, can't pee, can't climb a tree ": conjunctivitis/uveitis, urethritis, asymmetric oligoarthritis of lower limbs. Also enthesitis, dactylitis, keratoderma blennorrhagicum, circinate balanitis.	Synovial fluid inflammatory, culture-negative (sterile) . RF–, HLA-B27+. Stool/urine NAAT may show trigger (Chlamydia). X-ray: asymmetric sacroiliitis, periostitis.	Triad: arthritis + conjunctivitis + urethritis after diarrhea or STI. Sterile joint (vs septic/gonococcal). Formerly Reiter syndrome. Tx: NSAIDs; antibiotics for active Chlamydia.
Psoriatic arthritis STEP 1	Seronegative spondyloarthropathy in ~½ of psoriasis patients; HLA-B27 (axial) and HLA-Cw6; IL-17/IL-23 and TNF-driven. Joint disease may precede skin lesions.	Asymmetric oligoarthritis, DIP involvement, dactylitis ("sausage digits"), nail pitting/onycholysis , enthesitis, sacroiliitis; severe arthritis mutilans . Silvery scaly plaques on extensors.	X-ray: " pencil-in-cup " deformity of DIP, erosions with new bone formation, asymmetric sacroiliitis. RF and anti-CCP negative. ↑ ESR/CRP.	DIP + nail pitting + sausage digits + pencil-in-cup = psoriatic (contrast RA spares DIP; OA has Heberden without inflammation). Tx: methotrexate, anti-TNF, IL-17/IL-23 inhibitors.
Enteropathic arthritis IBD-associated arthritis STEP 1	Seronegative spondyloarthropathy with Crohn disease or ulcerative colitis ; gut inflammation → ↑ bacterial antigen exposure, IL-23/IL-17. Axial disease linked to HLA-B27 .	Peripheral: asymmetric large-joint (knee, ankle) oligoarthritis that parallels bowel activity . Axial: sacroiliitis/spondylitis independent of bowel activity . Also erythema nodosum, pyoderma gangrenosum, uveitis, PSC (UC).	RF–; HLA-B27+ in axial disease. X-ray: sacroiliitis, syndesmophytes like AS. Synovial fluid inflammatory, culture-negative. Colonoscopy/biopsy confirm IBD; ↑ ESR/CRP, ↑ fecal calprotectin.	Peripheral arthritis flares with colitis; axial disease does not. Part of PAIR . Trap: avoid NSAIDs in active IBD; anti-TNF (infliximab, adalimumab) treats both gut and joints.

High-yield patterns across the block

BONE TUMOR BY LOCATION

- Epiphysis**: giant cell tumor (adult, "soap bubble"), chondroblastoma.
- Metaphysis**: osteosarcoma, osteochondroma, aneurysmal bone cyst, non-ossifying fibroma.
- Diaphysis**: Ewing sarcoma (onion skin), osteoid osteoma (cortex), fibrous dysplasia.

TRANSLOCATIONS & GENES

- Ewing **t(11;22)** EWSR1-FLI1 · synovial sarcoma **t(X;18)** SS18-SSX.
- Myxoid liposarcoma **t(12;16)** FUS-DDIT3 · well-differentiated liposarcoma **MDM2** amplification.

- **Axial / pelvis, older adult:** chondrosarcoma. **Vertebrae:** osteoblastoma, metastases, myeloma.

CRYSTALS & JOINT FLUID

- **Gout:** needle-shaped, **negatively** birefringent, yellow when parallel; 1st MTP.
- **Pseudogout (CPPD):** rhomboid, weakly **positively** birefringent, blue when parallel; knee; chondrocalcinosis.
- WBC <2,000 noninflammatory · 2,000–50,000 inflammatory · >50,000 septic until proven otherwise.

- Alveolar rhabdomyosarcoma **t(2;13)** PAX3-FOXO1 · nodular fasciitis **MYH9-USP6**.
- Osteosarcoma **RB1, TP53** · fibrous dysplasia **GNAS** · achondroplasia **FGFR3** · OI **COL1A1/2**.

MUSCLE & NERVE BIOPSY PATTERNS

- **Myopathic:** rounded atrophic fibers, necrosis/regeneration, internal nuclei, fibrosis → fat.
- **Neurogenic:** angular atrophic fibers → fiber-type grouping → grouped atrophy.
- **Dermatomyositis** = perifascicular atrophy · **polymyositis** = endomysial CD8+ · **IBM** = rimmed vacuoles.
- **Axonal** = Wallerian degeneration, myelin ovoids · **demyelinating** = thin myelin, onion bulbs.